[Total No. of Question -04]

[Total No. of Pages - 01]

G H Raisoni International Skill Tech University, Pune

GH raisoni INTERNATIONAL SKILL TECH UNIVERSITY Pune

F.Y. B. Tech

TERM-I (Regular)

CAE I- WINTER 2025 (2024 Pattern)

Environmental Chemistry (25UBSL1102)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	What is hardness of water? Distinguish between temporary and permanent hardness of water.	[5]	1	1
b)	What is boiler corrosion? Explain its causes and preventions.	[5]	1	2
	OR			
Q 2.(a)	Explain the scale and sludge formation in boiler, its causes, and preventions.	[5]	1	2
(b)	A water sample contains 16.2 mg of $Ca(HCO_3)_2$, 36 mg of $MgSO_4$, 27.2 mg of $CaSO_4$, 2 mg of $MgCO_3$. Find the temporary, permanent, and total hardness of water. (Equivalent weights of $Ca(HCO_3)_2$=81, $MgSO_4$=60, $CaSO_4$=68, $MgCO_3$=42, $CaCO_3$=50)	[5]	1	3
Q 3.(a)	Explain solid waste with sources and effects. Elaborate all steps of solid waste management.	[5]	2	1
(b)	Explain solid waste management of your area with collection, segregation, transportation, and disposal.	[5]	2	3
	OR			
Q. 4 (a)	Elaborate sources of biological waste with its steps of management.	[5]	2	1
(b)	Explain principle of 3R's in reference of E-waste management. List two harmful effects of E-waste.	[5]	2	2